

Parvez Shaik

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Professional Summary

Software Engineer with experience in backend systems, AI retrieval, and cloud deployments. Worked on Node.js and Express.js services, PostgreSQL query tuning, React and TypeScript dashboards, and AWS EC2 deployments with Docker and Jenkins. Recent work includes Python based RAG and GraphRAG pipelines using LlamaIndex, SpaCy, FAISS, Redis, and MySQL. Strong in API development, performance tuning, deployment troubleshooting, and building systems that are reliable after release. Open to relocate across the U.S.

Technical Skills

Languages: Python, JavaScript, Java, TypeScript, C.

AI and LLM Retrieval: RAG, GraphRAG, PubMed RAG, LlamaIndex, FAISS, AutoGen, AG2, SpaCy, LLMs, Semantic Retrieval

Backend and API: Node.js, Express.js, Flask, FastAPI, Django, REST, GraphQL, gRPC, OAuth 2.0, JWT

Databases and Storage: SQL, PostgreSQL, MySQL, Redis, MongoDB, RocksDB

Cloud and DevOps: AWS EC2, AWS S3, AWS Lambda, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, Git, Terraform, Apache, Bash, CI/CD

ML and Data: TensorFlow, Pandas, Scikit-learn, RNN, Data Preprocessing, Model Evaluation

Frontend: React.js, Redux, TypeScript, Tailwind CSS, Lighthouse

Systems: Concurrency, Latency Optimization, Query Optimization, Idempotency, Observability, Root Cause Analysis

Experience

Software Developer Intern - MyEdMaster

Jun. 2025 - Aug. 2025

- Worked on a Python based AI content pipeline using MySQL, Redis, GraphRAG. Integrated AutoGen agents into the content review workflow. Reduced manual content evaluation time by 23%.
- Improved retrieval workflow for a proprietary LLM using LlamaIndex and SpaCy based on educator review feedback. Supported generation of 2,800 plus topic specific assessment items. Improved content relevancy from 72% to 83% using educator review metrics.
- Deployed a Flask backend on AWS EC2 using Docker and Apache. Worked with the backend team to troubleshoot 4 recurring deployment issues through root cause analysis. Improved model evaluation speed by 18%.

Software Development Engineer II - Cognizant

Sep. 2022 - Apr. 2024

- Partnered with backend engineers on Node.js and Express.js microservices. Tuned PostgreSQL indexing plus query execution. Reduced data retrieval latency from 420 ms to 295 ms.
- Led root cause analysis for recurring deployment failures with DevOps plus QA teams. Diagnosed failure patterns using Jenkins logs and Docker runtime checks on AWS EC2. Improved deployment reliability to 99.7%.
- Found sprint release bottlenecks in Jenkins and EC2 deploy logs. Added Docker, config, and smoke test checks in Jenkins. Cut release time by 35 minutes per sprint and reduced failures by 45 percent.
- Collaborated with UI/UX teams on React.js and TypeScript frontend modules. Used Redux for state management and Tailwind CSS for styling. Improved frontend responsiveness by 28%. Raised Lighthouse score from 78 to 91.

Software Development Engineer - Cognizant

Aug. 2021 - Sep. 2022

- Built React.js dashboards for internal analytics workflows using GraphQL and Django APIs. Supported near real time reporting for internal users.
- Improved slow refresh behavior in React dashboards during feature updates. Worked with backend engineers on data fetch handling plus component rendering and improved data rendering speed by 21.6%.
- Helped containerize backend services on AWS EC2 using Docker and Jenkins for deployment environments. Cut setup time per environment from 82 minutes to 39 minutes. Reduced manual deployment errors by 17%.

Software Development Intern - Widhya

Jan. 2021 - Jul. 2021

- Assisted in building an RNN based text classification model using TensorFlow and SpaCy. Refined tokenization and hyperparameter settings, which raised validation accuracy from 88.4% to 93.2%.
- Worked with the ML team on data preprocessing and model evaluation using Python (Pandas, Scikit learn).

Automated experiment scripts to reduce repetitive manual testing, cutting testing time by 27%.

Projects

Agentive Med, Multi Agent AI System for Alzheimer's Care | Python, AG2, FAISS, PubMed RAG, LLMs

- Built AG2 multi agent system using supervisor orchestration and PubMed backed FAISS RAG. Implemented tool triage and domain guardrails to cross verify LLM outputs against medical citations.
- Observed hallucinated medical advice in single agent QA responses during testing. Developed separate Nutrition plus Medical agents to isolate domain knowledge and used web search fallback for response verification against retrieved sources.

Distributed Vector Database for AI Applications | Python, gRPC, RocksDB, LlamaIndex, AWS S3, GraphRAG

- Designed distributed vector database using gRPC for node communication plus RocksDB as the storage engine. Integrated GraphRAG plus LlamaIndex for semantic retrieval across 140M plus embeddings.
- Hit 93 ms average query latency on large LLM retrieval workloads. Optimized indexing plus retrieval across distributed nodes for context aware search at scale.

Education

MS in Computer Science - Indiana University Bloomington

Aug. 2024 - May. 2026